

1. Show that if A is self-adjoint, then e^{tA} is contractive if all eigenvalues of A are nonpositive.

Solution:

Proof. Assume A is self-adjoint with nonpositive eigenvalues. Since A is self-adjoint, $A^T = \bar{A}$, so by the spectral theorem, A is diagonalizable by a matrix consisting of an orthonormal basis. For $A, n \times n$, let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of A with $\vec{u}_1, \dots, \vec{u}_n$ the corresponding eigenvectors. Recall that $\lambda_i \leq 0$ for all $i = 1, \dots, n$ and $\vec{u}_i \cdot \vec{u}_j = 0$ and $\|\vec{u}_i\| = 1$ for $i, j = 1, \dots, n (i \neq j)$. Then,

$$A = PDP^{-1}, \quad P = (\vec{u}_1 \quad \cdots \quad \vec{u}_n), \quad D = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}.$$

Since P forms a basis for $\mathbb{C}^n$ and $\vec{x} \in \mathbb{C}^n$ we have

$$\vec{x} = c_1 \vec{u}_1 + \cdots + c_n \vec{u}_n$$

for some $c_1, \dots, c_n \in \mathbb{C}$. Then,

$$e^{tA} \vec{x} = e^{t\lambda_1} c_1 \vec{u}_1 + \cdots + e^{t\lambda_n} c_n \vec{u}_n$$

so

$$\|\vec{x}\|^2 = c_1^2 + \cdots + c_n^2 \quad \text{and} \quad \|e^{tA} \vec{x}\|^2 = e^{t\lambda_1} c_1^2 + \cdots + e^{t\lambda_n} c_n^2.$$

Since $e^{t\lambda_i} c_i^2 \leq c_i^2$ because $\lambda_i \leq 0$ for $i = 1, \dots, n$, we have

$$\|\vec{x}\| \leq \|e^{tA} \vec{x}\|$$

for all t . □

2. Show that if K is a skew-adjoint square matrix, then e^{tK} is norm-preserving, that is, $\|e^{tK} x\| = \|x\|$ for all x and all $t \geq 0$. Furthermore, show that e^{tK} is unitary, that is, $(e^{tK})^{-1} = (e^{tK})^*$.

Solution:

Proof. Let K be a skew-adjoint matrix so $K^* = -K$, by definition. We first show that e^{tK} is unitary and then that it is norm-preserving. Since $K^* = -K$, we have

$$(e^{tK})^* = e^{tK^*} = e^{-tK}.$$

Also, K commutes with $-K$, so

$$e^{tK} e^{-tK} = e^{tK-tK} = e^0 = I.$$

Thus, e^{-tK} is the inverse of e^{tK} . It follows that

$$(e^{tK})^* = (e^{tK})^{-1}$$

so e^{tK} is unitary. Now, let $x \in \mathbb{C}^n$. Since e^{tK} is unitary, we have that

$$\begin{aligned} \|e^{tK} x\|^2 &= \langle e^{tK} x, e^{tK} x \rangle = \langle x, (e^{tK})^* e^{tK} x \rangle = \langle x, Ix \rangle = \langle x, x \rangle = \|x\|^2 \\ \Rightarrow \|e^{tK} x\| &= \|x\| \end{aligned}$$

thus, e^{tK} is norm-preserving for all $x \in \mathbb{C}^n$ and $t \geq 0$. □

3. Applying the decomposition $A = H + K$, use the Trotter product formula to show that e^{tA} is contractive if H has only non-positive real eigenvalues.

Solution:

Proof. Let $A = H + K$ for $H = \frac{1}{2}(A + A^*)$ and $K = \frac{1}{2}(A - A^*)$ such that H is self-adjoint and K is skew-adjoint. Assume as well that H has only non-positive eigenvalues. We will show that e^{tA} is contractive for every $t \geq 0$, or equivalently, $\|e^{tA}\| \leq 1$. Since H is self-adjoint and all of its eigenvalues are nonpositive, it follows from exercise 1 that e^{tH} is contractive so $\|e^{tH}\| \leq 1$ for all $t \geq 0$. By exercise 2, we have that e^{tK} is norm-preserving and thus unitary since K is skew-adjoint. It follows that $\|e^{tK}\| = 1$ for $t \geq 0$. Now, fix some $t \geq 0$ so by the Trotter Product ,

$$e^{t(H+K)} = \lim_{n \rightarrow \infty} \left(e^{\frac{t}{n}H} e^{\frac{t}{n}K} \right)^n \Rightarrow \|e^{t(H+K)}\| = \lim_{n \rightarrow \infty} \|(e^{\frac{t}{n}H} e^{\frac{t}{n}K})^n\|$$

by taking operator norms and its continuity over limits. Then,

$$\|(e^{\frac{t}{n}H} e^{\frac{t}{n}K})^n\| \leq \|e^{\frac{t}{n}H} e^{\frac{t}{n}K}\|^n \leq \left(\|e^{\frac{t}{n}H}\| \|e^{\frac{t}{n}K}\| \right)^n.$$

Since $\|e^{\frac{t}{n}H}\| \leq 1$ and $\|e^{\frac{t}{n}K}\| = 1$, we have that

$$\left(\|e^{\frac{t}{n}H}\| \|e^{\frac{t}{n}K}\| \right)^n \leq 1^n = 1$$

so $\|(e^{\frac{t}{n}H} e^{\frac{t}{n}K})^n\| \leq 1$. Passing the limit gives

$$\|e^{t(H+K)}\| \leq 1 \Rightarrow \|e^{tA}\| \leq 1$$

so e^{tA} is contractive. □

4. Suppose that A is any constant square matrix and that e^{tA} is contractive for all $t \geq 0$. Show then that A is dissipative.

Solution:

Proof. Assume that A is a constant square matrix and that e^{tA} is contractive for all $t \geq 0$. We must show that A is dissipative, or equivalently

$$\Re \langle x, Ax \rangle \leq 0$$

for all $x \in \mathbb{C}^n$. We start by fixing $x \in \mathbb{C}^n$ so that

$$\|e^{tA}x\| \leq \|x\| \Rightarrow \|e^{tA}\|^2 \leq \|x\|^2 \Leftrightarrow \langle e^{tA}x, e^{tA}x \rangle \leq \langle x, x \rangle$$

since e^{tA} is contractive. Now, subtractive $\langle x, x \rangle$ from both sides and dividing by t we see that

$$\frac{\|e^{tA}x\|^2 - \|x\|^2}{t} \leq 0.$$

Consider,

$$\begin{aligned} \|e^{tA}x\|^2 - \|x\|^2 &= \langle e^{tA}x - x, e^{tA}x \rangle + \langle x, e^{tA}x - x \rangle \\ \Rightarrow \frac{\|e^{tA}x\|^2 - \|x\|^2}{t} &= \left\langle \frac{e^{tA} - I}{t}x, e^{tA}x \right\rangle + \left\langle x, \frac{e^{tA} - I}{t}x \right\rangle. \end{aligned}$$

Now, taking the limit as $t \rightarrow 0^+$ and applying the facts that

$$\lim_{t \rightarrow 0^+} e^{tA}x = x \quad \text{and} \quad \lim_{t \rightarrow 0^+} \frac{e^{tA} - I}{t}x = Ax$$

we have that

$$\lim_{t \rightarrow 0^+} \frac{\|e^{tA}x\|^2 - \|x\|^2}{t} = \langle Ax, x \rangle + \langle x, Ax \rangle.$$

Since $\langle Ax, x \rangle = \overline{\langle x, Ax \rangle}$, it follows that

$$\langle Ax, x \rangle + \langle x, Ax \rangle = 2\Re\langle x, Ax \rangle.$$

On the other hand, because

$$\frac{\|e^{tA}x\|^2 - \|x\|^2}{t} \leq 0$$

for all $t > 0$, passing the limit gives $2\Re\langle x, Ax \rangle \leq 0$, hence, $\Re\langle x, Ax \rangle \leq 0$ for all $x \in \mathbb{C}^n$. Therefore, A is dissipative. $\square$

5. Consider

$$A_1 = \begin{pmatrix} -1/3 & 1 \\ 0 & -1/3 \end{pmatrix} \quad \text{and} \quad A_2 = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}.$$

Show that although A_1 has only negative eigenvalues, A_1 is not dissipative, and so e^{tA_1} is not contractive. Show that A_2 is dissipative, and so e^{tA_2} is contractive.

Solution:

Proof. Consider

$$A_1 = \begin{pmatrix} -1/3 & 1 \\ 0 & -1/3 \end{pmatrix} \quad \text{and} \quad A_2 = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}.$$

We will show that A_1 has only negative eigenvalues but is not dissipative while A_2 is. Since A_1 is an upper triangular matrix, we know the eigenvalues of A_1 are $\lambda_1 = \lambda_2 = -1/3$, and so, A_1 has only negative eigenvalues. Now let

$$H_1 = \frac{1}{2}(A_1 + A_1^*) = \frac{1}{2} \begin{pmatrix} -1/3 & 1 \\ 0 & -1/3 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} -1/3 & 0 \\ 1 & -1/3 \end{pmatrix} = \begin{pmatrix} -1/3 & 1/2 \\ 1/2 & -1/3 \end{pmatrix}$$

such that H_1 is self-adjoint by definition. Then,

$$\det H_1 = \left(\frac{-1}{3}\right)^2 - \left(\frac{1}{2}\right)^2 = \frac{1}{9} - \frac{1}{4} < 0$$

so the eigenvalues of H_1 have opposite signs, or equivalently, H_1 has one positive eigenvalue. It follows that there exists $x \in \mathbb{C}^2$ such that $\langle x, H_1 x \rangle > 0$, but

$$\Re\langle x, Ax \rangle = \left\langle x, \frac{1}{2}(A_1 + A_1^*)x \right\rangle = \langle x, H_1 x \rangle > 0.$$

From exercise 4, if e^{tA_1} were contractive for all $t \geq 0$, then A_1 would have to be dissipative. Since A_1 is not dissipative, it follows that e^{tA_1} is not contractive. Now consider A_2 and define

$$H_2 = \frac{1}{2}(A_2 + A_2^*) = \begin{pmatrix} -1 & 1/2 \\ 1/2 & -1 \end{pmatrix}$$

such that H_2 is also self-adjoint by definition. The eigenvalues of H_2 are given by

$$\lambda^2 + 2\lambda + \frac{3}{4} = \left(\lambda + \frac{1}{2}\right) \left(\lambda + \frac{3}{4}\right) = 0 \Rightarrow \lambda_1 = -\frac{1}{2}, \lambda_2 = -\frac{3}{4}.$$

Since $\lambda_1, \lambda_2 \leq 0$, H_2 is self-adjoint with non-positive eigenvalues so

$$\langle x, H_2 x \rangle \leq 0 \Rightarrow \Re \langle x, A_2 x \rangle = \langle x, H_2 x \rangle \leq 0$$

for all $x \in \mathbb{C}^2$. It follows that A_2 is dissipative. Then, by exercise 3, e^{tA_2} is contractive for all $t \geq 0$ since H_2 has eigenvalues all nonpositive. $\square$

⁰LaTeX code for this document can be found on github [here](#)